

Daily Review



1 Name the four types of engineer.

Electrical, mechanical,
civil, chemical

2 Who designs a bridge's foundations?

Civil

3 What is a prototype?

A first version, built to
test

4 Order: build, improve, design, test

Design, build, test,
improve

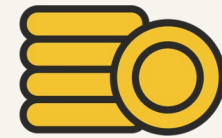
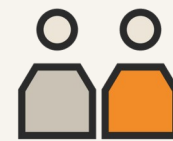
5 A chemical engineer's job, and why?

A coating — **so no rust**

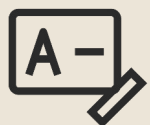
Impacts of Engineering

Who gains, who loses — and what happens to the planet?

S3 Engineering Science



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

The Borders Railway reopened in 2015. Who gained — and did anyone lose?



Borders Railway

PART 1

Impacts of Engineering

Good and bad effects on people, money and the natural world.

By the end of this part...



Rate each one 1-3: 1 not yet, 2 nearly, 3 got it.

- 1 I can describe positive and negative effects on society
- 2 I can describe positive and negative economic effects
- 3 I can describe positive and negative environmental effects
- 4 I can use real examples in my answers
- 5 I can explain my ideas clearly using what + why

SECTION A

Three Kinds of Impact

More jobs, cleaner air, safer streets — the same kind of effect?

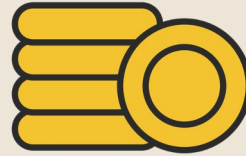
The Three Types of Impact

Impact: an effect or change caused by something.



Social

people's lives:
safety, travel,
health



Economic

money and jobs:
costs, profits,
tourism



Environmental

the natural world:
pollution, CO₂,
habitats, waste

Worked Example – Sort the Borders Railway

Social

easier trips to work or college in Edinburgh

Economic

tourists spend money in Borders shops

Environmental

fewer car trips, so less CO₂

Trap

“saves money” is always economic



Show Me — Which Type of Impact?



1

Faster broadband lets pupils do homework online.

Social

2

A new ski centre brings tourists who spend in local shops.

Economic

3

A new reservoir floods a valley where deer lived.

Environmental

Show Me — Which Type of Impact?



4

LED street lights cut the council's electricity bill.

Economic

5

Electric cars give out no exhaust in a town centre.

Environmental

Show Me — Which Type of Impact?

EXTRA PRACTICE



1

Self-checkouts mean fewer checkout jobs.

Economic

2

A new footbridge: children no longer cross a busy road.

Social

3

Old phones thrown away fill up landfill.

Environmental

Show Me — Which Type of Impact?

EXTRA PRACTICE



4

The LED lights also mean less electricity from power stations.

Environmental

5

Why can one project have all three types of effect?

It changes people, money and nature

SECTION B

Good and Bad

Is a wind farm good or bad? It depends who you ask.

Both Sides — What, Why and Who

What

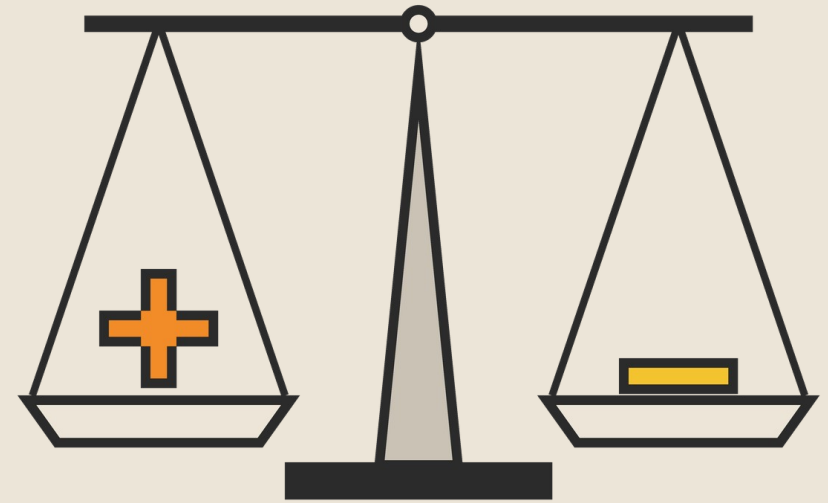
the effect itself

Why

why it happens, or why it matters

Who

who or what is affected



Most projects have good
AND bad effects.

Worked Example – Smartphones

Describe one positive and one negative social effect. (2 marks)

Positive

can call for help anywhere, so safer out alone

Negative

phone use late at night, so less sleep

Scores 0

“they're useful” – no what, no why



Show Me — Self-Checkouts



1

Positive economic effect for the supermarket

Fewer staff — so saves money

2

Negative economic effect for staff

Fewer till jobs — so some lose work

3

Positive social effect for shoppers

Shorter queues — so a quicker trip

Show Me — Self-Checkouts



4 Negative social effect for shoppers

Screens confuse some
— **so they need help**

5 Which scores: A “bad for jobs” or B
“fewer till staff, so some lose jobs”?

B — it says what AND
why

Show Me — Online Gaming

EXTRA PRACTICE



- 1 A positive social effect
- 2 A negative social effect
- 3 An economic effect for a games company

Play with friends far away — **so less lonely**

Hours online — **so less sleep or exercise**

Sells games — **so makes a profit**

Show Me — Online Gaming

EXTRA PRACTICE



4

An environmental effect when old consoles are thrown away

Electronic waste in
landfill

5

Improve: “gaming is bad for you”.

Late gaming — so less
sleep

SECTION C

Real Examples

“A wind farm is good for the environment.” Which one?
Where?

Four Scottish Projects (1 of 2)



Whitelee Wind Farm

electricity with no fuel
burned, so less CO₂



Pitlochry Dam

renewable electricity — but it
blocks salmon

Four Scottish Projects (2 of 2)



Queensferry Crossing

wind shields keep it open, so
journeys are reliable



Borders Railway

easier travel to Edinburgh;
tourists visit

Worked Example – Build Up an Answer

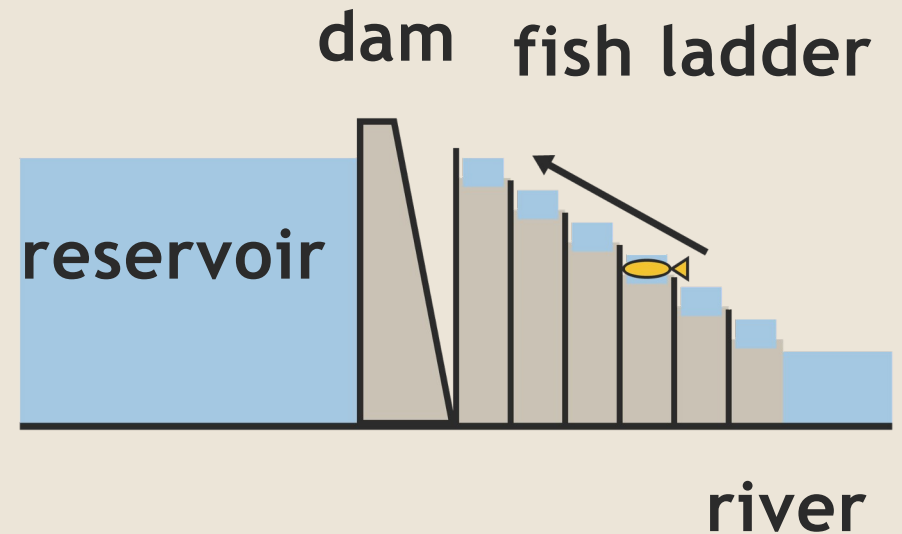
Describe one negative environmental effect of a hydro dam.

Scores 0

“dams are bad for fish”

What + why

the dam blocks the river, so salmon can't swim up to breed



Worked Example — Add a Real Example

Describe one negative environmental effect of a hydro dam.

What + why

the dam blocks the river, so salmon can't swim up to breed

+ real example

Pitlochry Dam has a fish ladder — a staircase of pools salmon can climb



Show Me — Real Examples



1

Name one real renewable-energy project in Scotland.

Whitelee Wind Farm,
Pitlochry Dam

2

Queensferry Crossing: one social effect, what + why

Stays open — **so people
get to work**

3

Whitelee: one positive environmental effect, what + why

No fuel burned — **so
less CO₂**

Show Me — Real Examples



4

Why does Pitlochry Dam have a fish ladder?

So salmon can get past the dam to breed

5

A real engineering project near you — and one impact.

Your own answer

Show Me — Real Examples

EXTRA PRACTICE



1

Borders Railway: who gains money, and why?

Borders shops —
tourists come by train

2

One negative effect of building the Borders Railway

It cost a lot to build

3

Improve with a real example: “wind farms help the environment”.

Whitelee — so less CO₂

Show Me — Real Examples

EXTRA PRACTICE



4

Can one effect be good for one group and bad for another?

Yes — self-checkouts:
shop gains, staff lose

5

Why does naming a real project make an answer stronger?

It proves the effect
really happens

“A new bus route means people can get to the hospital more easily.” Which type?



- A economic
- B social
- C environmental
- D it isn't an impact

Answer B: It changes people's lives — getting to a service is social.

“More tourists visit, so local cafés take more money.”
Which type?



- A social
- B environmental
- C economic
- D all three equally

Answer C: The effect is money coming in – that is economic.

Which is an environmental effect of solar panels on a school roof?



- A the school saves money on bills
- B pupils learn about energy
- C the panels cost a lot to fit
- D less fuel burned, so less CO₂

Answer D: Saving money is economic, even on a green project.

Can one project have both positive and negative effects?



- A yes — most projects have both
- B no — it's either good or bad
- C only if something goes wrong
- D only very large projects

Answer A: Almost every project helps someone and costs something.

“Describe one negative environmental effect of a new wind farm.” Which answer gets the mark?



- A “it's bad for the environment”
- B “it costs a lot to build”
- C “birds can be killed by the blades”
- D “wind turbines”

Answer C: Only C names a real effect on nature. B is economic.

Mini Test 2



Mini Test 2 sheet

On your own, in silence, pen only.

10 minutes – 8 marks.

Then swap and mark from the next slides.

Mini Test 2 — Marking

ANSWERS

Q1 (1 + 1 marks)

+ quieter, no exhaust, so cleaner air for people. – fewer routes while buses recharge, or disruption.

Scores 0

a money answer in Q1 — money is Q2

Mini Test 2 – Marking

ANSWERS

Q2 (1 + 1 marks)

+ electricity costs less than diesel, or jobs fitting chargers. – new buses and chargers are expensive, or diesel mechanics lose work.

Q3(a) (1 mark)

No exhaust from the buses, so less air pollution in town.

Mini Test 2 – Marking

ANSWERS

Q3(b) (1 mark)

Mining metals for batteries damages land, or old buses and batteries become waste.

Q4 (1 + 1 marks)

A named real project with a true impact (e.g. Whitelee: no fuel burned, so less CO₂) + the right type.

Scores 0

“better for the environment” on its own

Brain Dump



In your jotter

Write everything you remember about the impacts of engineering.
On your own, 2 minutes. No notes.
Then we check it against the Plenary.

Plenary – Good and Bad Effects



1

What does an economic impact affect?

Money and jobs

2

One negative environmental effect of building work.

Habitats damaged, or waste

3

One positive and one negative social effect of smartphones.

Safer when out – but less sleep

Plenary – Good and Bad Effects



4

A real project and one impact, what + why.

Whitelee – so less CO₂

5

The three things a strong impact answer says.

What, why and who

Plenary – Good and Bad Effects

EXTRA PRACTICE



1 “Staff lose their jobs” – which type?

Economic

2 Improve: “it's good for the environment”.

Say what changes in nature, and why

3 What problem does a fish ladder solve?

Salmon can't pass the dam

Plenary – Good and Bad Effects

EXTRA PRACTICE



4

“A new road cuts a journey to 10 minutes.” Which type?

Social

5

One real Scottish project from today.

e.g. Whitelee Wind
Farm

Look back — by the end of this part...



Rate each one 1-3 again. Compare with the start.

- 1 I can describe positive and negative effects on society
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LOOKING AHEAD

Lesson 3 — Energy

Forms of energy, transfers, and useful and wasted energy.

Credits

The big question (Borders Railway): © Russel Wills, CC BY-SA 2.0, via Geograph

Worked Example – Sort the Borders Railway:
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